Historical Patent Dates and AI Evidence Assessment

A document-evaluation pilot with supplementary US-France archival research

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Electrical and electronics engineering background

Executive finding

The US specification for Edison's patent 446,668 cites French patent 148,852 as dated 16 April 1883. France's INPI archive explicitly records 9 May 1882 as the initial patent's filing date. A screenshot of the original French document contains the same deposit date. The US-cited date must therefore not automatically be labelled the French filing date. Its precise event meaning remains unresolved. [1-3]

Evaluation result

Two researcher-supplied responses, obtained through Gemini in Chrome and Perplexity, each received 10/10 on five document-based questions under the pilot rubric. No core factual failure was identified. These were interface-level runs with undisclosed underlying model versions, not an overall model ranking. The French archival investigation was not part of either model test.

Scope and contribution

The researcher located records, extracted evidence, compared scans with searchable text, navigated an international archive and assessed responses. AI mentoring supported search strategy, task design, evidence review, scoring and documentation. This is an exploratory portfolio case study, not peer-reviewed research or an official GAIA/SimpleQA contribution.

Repository: github.com/sharlyne-agini/ai-research-benchmark-portfolio
Public accessibility must be checked separately.

Question and method

Can historical electrical-lighting patent references be checked across national records without confusing filing dates with other patent dates? The researcher inspected US446668, extracted dates from PDF page 2, compared searchable transcription against the scan, and followed foreign-patent references. British-record searches were unsuccessful. A patent-number search in INPI located initial and addition records for French patent 148852.

Record	Evidence established	Location
US446668	Application filed 7 August 1882; serial 68,651	PDF p. 2
US446668	Letters Patent dated 17 February 1891	PDF p. 2; drawing p. 1
US446668	French 148,852 associated with 16 April 1883	PDF p. 2
INPI 1BB148852	Initial filing: 9 May 1882; applicant EDISON Thomas-Alva	Catalogue capture
INPI 1BB148852_1	Addition filing: 26 December 1882	Catalogue capture

The original French first-image screenshot contains the patent number and a 9 May 1882 deposit passage. It corroborates the catalogue entry; it is not an independent institution or second human verification. The initial record concerns improvements to electric arc lighting and regulating mechanisms. [2-4]

Interpretation

The same patent number identifies an initial record and an addition with different filing dates. Record type and archive reference matter. Different dates may describe different events; the evidence does not establish that either national record is erroneous. Identical technical claims or legal scope across countries were not established.

Unresolved boundary

The French publication date and meaning of 16 April 1883 remain undetermined. The British patent was not independently retrieved. These are declared limitations, not completed verification claims. The case closes on the narrower, supported finding about date labels.

Procedure and questions

Each run used only 00446668.pdf and equivalent instructions: identify (1) US number, title and inventor; (2) filing date and serial; (3) Letters Patent date; (4) British number and associated date with its wording; and (5) whether separate British filing/publication dates are established. The prompt requested PDF pages and supporting wording, distinguished statement from inference, and prohibited web search.

Condition	Gemini	Perplexity
Test date	21 October 2026	21 October 2026
Interface / mode	Chrome side panel / Auto	Web / Learn step by step
Underlying model	Not disclosed	Not disclosed
New chat; no answer key	Researcher-confirmed	Researcher-confirmed
Actual tool use	Not independently verified	Not independently verified

Gemini's interface also displayed sharing of the New Tab page. Perplexity used equivalent question wording without numbering. Differences in interface context and configuration limit comparability.

Question	Gemini	Perplexity
1. Patent identification	2/2	2/2
2. Filing date and serial	2/2	2/2
3. US Letters Patent date	2/2	2/2
4. British reference and wording	2/2	2/2
5. Evidence limits	2/2	2/2
Total	10/10	10/10

Rubric: 2 = correct and complete with accurate page/evidence support and bounded conclusions; 1 = substantively correct but incomplete or inadequately referenced; 0 = incorrect, absent or materially unsupported. Researcher accepted the scores with AI assistance. The rubric followed the Gemini response and preceded Perplexity, making the first run exploratory.

Both responses correctly avoided identifying the British associated date as a separately established filing or publication date. Their extra claim that PDF page 1 shows the US patented date has now been visually checked and supported. Character-for-character quotation fidelity was not fully audited. No hidden reasoning was available.

Reproduction and audit trail

Use the source PDF and preserve the original prompts and responses in tests/pilot-001-gemini.md and tests/pilot-002-perplexity.md. Use pilot-rubric.md for grading. For a repeat: select and record an exact model identifier where available, start a fresh conversation, attach only the PDF, withhold the answer key, record settings and tool access, preserve the entire response and score each question consistently. Treat repeats as new runs.

Limitations

One document, five related questions and one run per interface cannot support statistical or general performance claims. Exact model versions and actual web-tool use were unknown; modes and prompt numbering differed. Source verification was researcher-led with AI review, without a second independent human. Copied response text is the result record; setup screenshots do not capture the full returned answers. There was no international-model evaluation, demonstrated red-teaming failure or official benchmark contribution.

References and evidence locations

1. US primary document
Thomas A. Edison, Electric-Arc Light, US446668. Rutgers-hosted scan, PDF pp. 1-2.

Open source

2. French initial record
INPI historical patents, archive reference 1BB148852, patent 148852. Catalogue filing date 09.05.1882.

Open source

3. French original image
INPI 1BB148852, viewer image 1 of 17. Supplied screenshot: french_patent_first_page.PNG.

4. French addition record
INPI archive reference 1BB148852_1, Certificate of Addition. Catalogue filing date 26.12.1882. Search patent 148852 and select the addition.

Open source

Archive evidence was inspected through researcher-supplied screenshots and saved HTML from 21 October 2026. Live INPI access may require interactive searching. Evidence folder: projects/ENG-001-historical-patents/evidence/. Screenshots corroborate fields visible at capture; they do not reproduce every page of the archive files.

Source PDF SHA-256
97fdd753fd97cae02b05ca7c908b76f4
ac98d7951f1c27d891473eb58d42af30

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